

James Boutros

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EDUCATION

University of Waterloo

BASc in Electrical and Computer Engineering

Assumption Catholic S.S.

Ontario Secondary School Diploma -- 98% Average, 4.0 Unweighted GPA

Sept. 2026 -- Present

Waterloo, ON

Graduated 2026

Burlington, ON

EXPERIENCE

Phydata -- Co-Founder and CTO

5-Person Founding Team

June 2026 -- Present

Toronto, CA (Remote)

- Co-founded a data-infrastructure startup building collection, labeling, and dataset-management tools for physical AI; secured \$20,000 in early investment and applied to Y Combinator W27.
- Designed and built the company's website (React, Python, Supabase) and wrote the YC application working document, covering market analysis, competitive landscape, and business-model phasing.
- Proposed a product-roadmap expansion to train proprietary foundation models on collected robot data; also built an automated social-media agent for the company's marketing.

Independent Machine Learning Researcher

Collaborative Research Team

June 2026 -- Present

Toronto, CA (Remote)

- Conducting research with collaborators to determine whether SHAP-based explanation drift improves the early detection of individual model errors under real-world distribution shifts, evaluating Logistic Regression, XGBoost, and TabPFN.
- Completed two studies on real datasets (hospital readmission and U.S. Census housing) using AUROC, AUPRC, and calibration metrics; found that the effect is architecture-dependent rather than universal, with the research progressing toward publication.

iOS Application Developer

Career Education Council / Apple Co-op

June 2025 -- July 2025

Toronto, CA (Remote)

- Designed and independently built MicroLoop, a full SwiftUI productivity app with micro-habit tracking, AI-powered goal breakdown, and streak gamification; presented the completed app to a panel of Apple engineers.
- Researched emerging iOS frameworks and maintained development documentation for team knowledge transfer.

Peer Tutor

Independent Contract

2025 -- 2026

Burlington, ON

- Delivered one-on-one tutoring in Advanced Functions, Chemistry, and Physics; designed personalized curriculum plans that helped students raise their average grades by over 15%.

PROJECTS

Jarvis -- Personal AI Operating System | Python, TypeScript, React, Obsidian, Kokoro, Playwright

- Built a voice-enabled personal AI operating system with an Obsidian-based knowledge brain for persistent context and connected personal workflows.
- Implemented tool-driven control of Gmail, Calendar, and web browsing through natural-language requests, with progressive text-to-speech playback and deterministic intent routing.

Morning Brief -- Personal Research & Life-Operations Agent | PowerShell, Gmail API, Google Calendar, Obsidian

- Built a daily agentic research-and-action system synthesizing academic deadlines, AI and hardware news, co-op signals, and personal priorities into a single, focused daily brief.
- Orchestrated a validator-first pipeline (research, source audit, Gmail-ready brief, duplicate-safe Calendar updates, Obsidian sync) backed by 84 supporting artifacts and anti-hallucination safeguards.

Embedded Flight & Security Simulations | ESP32, Arduino/C++, Wokwi, SimulIDE

- Spacecraft Systems Simulator: designed a distributed flight-control system across four microcontrollers with a full mission-state machine and serial-message protocols for telemetry and commands; built safety logic blocking arming on low battery or overheating and ported the simulation to SimulIDE using four Arduino Mega boards.
- ESP32 Vault Escape-Room Console: designed an interactive security console with a multi-stage state machine requiring proximity, light, motion, and PIN challenges to unlock.

TinkerCAD Embedded Systems Suite | C++, Arduino Uno, TinkerCAD

- Built four hardware systems: a dual-sensor fire alarm, a PIN-based two-door lock, an ISR-based pedestrian traffic light, and a photoresistor ADC light classifier.

TECHNICAL SKILLS

Languages: Swift, Python, C++, JavaScript/TypeScript, HTML/CSS

Frameworks: SwiftUI, Next.js, React, Framer Motion, Three.js / React Three Fiber

Tools: Arduino/ESP32, Git, Vercel, Supabase

Concepts: OOP, distributed systems, state machines, interrupt-driven I/O, REST APIs, model evaluation under distribution shift

EXTRACURRICULARS & AWARDS

Peer Tutoring -- Math, Chemistry, Physics

VEX Robotics Club -- Competed across multiple seasons, building several robots for regional competitions (Gr. 10--12)

Competitive Weightlifting -- Competed for three years

Completed 160+ hours of community volunteer service

2x Hackathon Winner

AP Calculus AB Exam -- Score of 5

Galois Waterloo Math Contest -- Top Score at School